

Homework 1 (60 points)

Due: 18/10, Fri, 11:59pm.

No late submission is allowed. Please upload your homework to Canvas. You can discuss with your peers, but you need to write your own copy.

Let $(\Omega, \mathcal{F}, \mathbb{P})$ is the probability space, and $(\Omega, \mathcal{F}, \mu)$ is a measurable space. Questions are testing: Borel-Cantelli Lemma, probabilistic inequalities and convergence theorems.

1. If X_n is any sequence of random variables, there are constants $c_n \rightarrow \infty$ so that $X_n/c_n \rightarrow 0$ a.s.

Define A_n s.t. $\mathbb{P}\left(|X_n| > \frac{|c_n|}{n}\right) \leq \frac{1}{2^n}$. Then, we have $\mathbb{P}\left(\left|\frac{X_n}{c_n}\right| > \frac{1}{n}\right) \leq \frac{1}{2^n}$. Then, according to Borel-Cantelli Lemma, we have $\sum_{n=1}^{\infty} \mathbb{P}\left(\left|\frac{X_n}{c_n}\right| > \frac{1}{n}\right) < \infty$ and $\mathbb{P}\left(\left|\frac{X_n}{c_n}\right| > \frac{1}{n}, \text{ i.o.}\right) = 0$. We can choose $N > 0$ s.t. $\varepsilon < \frac{1}{N}$. Then, we can obtain that $\mathbb{P}\left(\left|\frac{X_n}{c_n}\right| > \varepsilon, \text{ i.o.}\right) \leq \mathbb{P}\left(\left|\frac{X_n}{c_n}\right| > \frac{1}{N}, \text{ i.o.}\right) = 0$, then $X_n/c_n \rightarrow 0$ a.s.

2. Let $X_1, X_2, \dots$ be independent. Show that $\sup X_n < \infty$ a.s. if and only if $\sum_n \mathbb{P}(X_n > A) < \infty$ for some A .

If $\sup x_n < \infty$ a.s., we can suppose $\sum \mathbb{P}(X_n > A) = \infty$. Due to $X_1 \dots x_n$ are independent, according to Borel-cantelli Lemma, we have $\mathbb{P}(X_n > A \text{ i.o.}) = 1$ for all A . It means that $\sup X_n = \infty$ a.s. which contradicts $\sup X_n < \infty$ a.s.

If $\sum \mathbb{P}(X_n > A) < \infty$, similarly, we can have $\mathbb{P}(X_n > A \text{ i.o.}) = 0 \Rightarrow \mathbb{P}(X_n \leq A \text{ eventually}) = 1$. Therefore $\sup X_n < \infty$ a.s..

3. If $X_1, X_2, \dots$ are i.i.d. with $\mathbb{E}|X_i| = \infty$, then $\mathbb{P}(|X_n| \geq n \text{ i.o.}) = 1$. So if $S_n = X_1 + \dots + X_n$ then

$$\mathbb{P}\left(\lim_{n \rightarrow \infty} S_n/n \in (-\infty, \infty)\right) = 0.$$

We have

$$\mathbb{E}|X_1| = \int_0^{\infty} \mathbb{P}(|X_1| > x) dx \leq \sum_{n=0}^{\infty} \mathbb{P}(|X_1| > n)$$

Since $\mathbb{E}|X_1| = \infty$ and $X_1, X_2, \dots$ are i.i.d., it follows from the second Borel-Cantelli lemma that $\mathbb{P}(|X_n| \geq n \text{ i.o.}) = 1$. To prove the second claim, observe that

$$\frac{S_n}{n} - \frac{S_{n+1}}{n+1} = \frac{S_n}{n(n+1)} - \frac{X_{n+1}}{n+1}$$

and on $C = \{\omega : \lim_{n \rightarrow \infty} S_n/n \text{ exists} \in (-\infty, \infty)\}, S_n/(n(n+1)) \rightarrow 0$. So, on $C \cap \{\omega : |X_n| \geq n \text{ i.o.}\}$, we have

$$\left| \frac{S_n}{n} - \frac{S_{n+1}}{n+1} \right| > 2/3 \quad \text{i.o.}$$

contradicting the fact that $\omega \in C$. From the last observation, we conclude that

$$\{\omega : |X_n| \geq n \text{ i.o.}\} \cap C = \emptyset$$

and since $\mathbb{P}(|X_n| \geq n \text{ i.o.}) = 1$, it follows that $\mathbb{P}(C) = 0$. It also showed that $\mathbb{E}[|X|_1] < \infty$ is necessary for SLLN.

4. Let $X_1, X_2, \dots$ be independent with $\mathbb{P}(X_n = 1) = p_n$ and $\mathbb{P}(X_n = 0) = 1 - p_n$. Show that
- (i) $X_n \rightarrow 0$ in probability if and only if $p_n \rightarrow 0$, and
 - (ii) $X_n \rightarrow 0$ a.s. if and only if $\sum p_n < \infty$.

- (i) For $0 < \varepsilon < 1$, $\mathbb{P}(|X_n| > \varepsilon) = p_n \rightarrow 0 \iff p_n \rightarrow 0$.
- (ii) Let $A_n = \{|X_n| > \varepsilon\}$. Then, $\mathbb{P}(A_n \text{ i.o.}) = 0 \iff \sum_{n=1}^{\infty} \mathbb{P}(A_n) = \sum_{n=1}^{\infty} p_n < \infty$. Note that $(\Rightarrow)$ holds since X_n 's are independent.

5. Let $(X_1, X_2, \dots)$ be a sequence of independent real random variables, each with the distribution $\mathbb{P}(X_n > x) = e^{-x}$ when $x > 0$ and $\mathbb{P}(X_n > x) = 1$ when $x \leq 0$. Let

$$L = \limsup_n \frac{X_n}{\ln n},$$

then $\mathbb{P}(L = 1) = 1$.

Let A_n be the event that $X_n \geq \alpha \ln n$. Then

$$\mathbb{P}(A_n) = n^{-\alpha},$$

and the events $(A_1, A_2, \dots)$ are independent. Also, note that

$$\sum_n \mathbb{P}(A_n) \begin{cases} = \infty & \text{if } \alpha \leq 1 \\ < \infty & \text{if } \alpha > 1 \end{cases}$$

Thus, from the Borel-Cantelli Lemmas it follows that

$$\mathbb{P}(X_n \geq \alpha \ln n, \text{ i.o.}) = \begin{cases} 1 & \text{if } \alpha \leq 1 \\ 0 & \text{if } \alpha > 1 \end{cases}$$

Now, note that the event $\{L \geq \alpha\}$ is identical to the event

$$\cap_{m>0} \{X_n \geq (\alpha - 1/m) \ln n, \text{ i.o.}\}$$

and so $\mathbb{P}(L \geq 1) = 1$. Then it follows that $\mathbb{P}(L \geq 1 + 1/n) = 0$ for any $n > 0$, and so we have that $\mathbb{P}(L > 1) = 0$, hence, $\mathbb{P}(L \leq 1) = 1$, and so $\mathbb{P}(L = 1) = 1$.

6. We introduce convergence in probability, if we replace the probability measure by a general measure, in the measure space $(\Omega, \mathcal{F}, \mu)$, we say the random variable sequence $\{X_n\}$ convergence in measure if there exists a random variable X , it holds

$$\lim_{n \rightarrow \infty} \mu(\{|X_n(\omega) - X(\omega)| > \varepsilon\}) = 0, \quad \text{for any } \varepsilon > 0.$$

Suppose $f_n \rightarrow f$ in measure and $g_n \rightarrow g$ in measure.

(i) $f_n + g_n \rightarrow f + g$ in measure.

(ii) $f_n g_n \rightarrow f g$ in measure if $\mu(X) < \infty$, but not necessarily if $\mu(X) = \infty$.

(i) $\mu(|f_n + g_n - f - g| > \varepsilon) \leq \mu(|f_n - f| > \varepsilon) + \mu(|g_n - g| > \varepsilon) \rightarrow 0$ as $n \rightarrow \infty$.

(ii)

$$\begin{aligned} \mu(|X_n Y_n - X Y| > \varepsilon) &\leq \mu\left(|X| |Y_n - Y| \geq \frac{1}{2} \varepsilon\right) + \mu\left(|Y_n| |X_n - X| \geq \frac{1}{2} \varepsilon\right) \\ \mu\left(|X| |Y_n - Y| \geq \frac{1}{2} \varepsilon\right) &= \mu\left(|X| |Y_n - Y| \geq \frac{\varepsilon}{2}, |X| \leq M_1\right) + \mu\left(|X| |Y_n - Y| \geq \frac{\varepsilon}{2}, |X| > M_1\right) \\ &\leq \mu\left(|Y_n - Y| > \frac{\varepsilon}{2M}\right) + \mu(|X| > M_1) \\ \mu\left(|Y_n| |X_n - X| \geq \frac{\varepsilon}{2}\right) &= \mu\left(|X_n - X| |Y_n| > \frac{1}{2} \varepsilon, |Y_n - Y| \geq 1\right) + \mu\left(|X_n - X| |Y_n| \geq \frac{\varepsilon}{2}, |Y_n - Y| < 1\right) \\ &\leq \mu(|Y_n - Y| \geq 1) + \mu\left(|X_n - X| (1 + |Y_1|) \geq \frac{\varepsilon}{2}\right) \\ &\leq \mu(|Y_n - Y| \geq 1) + \mu\left(|X_n - X| \geq \frac{\varepsilon}{2(1 + M_2)}\right) + \mu(|Y| > M_2) \end{aligned}$$

Due to $\mu(x) < \infty$, therefore, as $\mu_1, \mu_2 \rightarrow \infty$, we can have $X_n Y_n \xrightarrow{p} X Y$.

7. Let $g > 0$ almost everywhere, and the inequality holds

$$\left(\int_0^1 x g(x) dx\right)^3 \leq \frac{4}{25} \int_0^1 g^3(x) dx.$$

Finding all functions $\int_0^1 |g(x)|^3 dx < \infty$ holding the equality.

$g > 0$ a.e. and $x \in (0, 1)$

$$\begin{aligned} \int_0^1 xg(x)dx &\leq \left(\int_0^1 x^{\frac{3}{2}}dx\right)^{\frac{2}{3}} \cdot \left(\int_0^1 g^3(x)dx\right)^{\frac{1}{3}} \\ \Rightarrow \left(\int_0^1 xg(x)dx\right)^3 &\leq \left(\int_0^1 x^{\frac{3}{2}}dx\right)^2 \cdot \left(\int_0^1 g^3(x)dx\right) = \frac{4}{25} \cdot \left(\int_0^1 g^3(x)dx\right). \end{aligned}$$

To hold the equality, we only need

$$x^{3/2} = cg(x)^3, \quad c \text{ is a constant s.t. } c \neq 0.$$

8. Let X be an integrable function on $(\Omega, \mathcal{F}, \mu)$. Then for any $\varepsilon > 0$, there exists $\delta > 0$ such that

$$F \in \mathcal{F}, \mu(F) < \delta \implies \int_F |X|d\mu < \varepsilon$$

Due to X is an integrable function, there is a simple function ϕ s.t. $\int f - \phi d\mu < \frac{\varepsilon}{2}$. We can rewrite $\phi = \sum a_k I_{A_k} \Rightarrow \int_A \phi d\mu = \sum a_k \mu(A_k \cap A) \leq \mu(A) \sum_{k=1}^n a_k$ we can set $\delta = \frac{\varepsilon}{2\sum a_k}$. Then $\int_F |x|d\mu \leq \int_F |x - \phi|d\mu + \int_F |\phi|d\mu < \frac{\varepsilon}{2} + \mu(F) \sum a_k < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$.

or

Let $X_n \triangleq |X|1_{\{|X|>n\}}$. By DCT, $\int_\Omega X_n d\mu = \int_{\{|X|>n\}} |X|d\mu \rightarrow 0$ as $n \rightarrow \infty$, hence, for any $\varepsilon > 0$, there exists $n > 0$ such that $\int_{\{|X|>n\}} |X|d\mu < \frac{\varepsilon}{2}$. For any measurable set A , we compute

$$\begin{aligned} \int_A |X|d\mu &= \int_{F \cap \{|X| \leq n\}} |X|d\mu + \int_{A \cap \{|X| > n\}} |X|d\mu \\ &\leq n\mu(A) + \int_{\{|X| > n\}} |X|d\mu < n\mu(A) + \frac{\varepsilon}{2} < \varepsilon, \quad \delta = \frac{\varepsilon}{2n}. \end{aligned}$$

9. If $\int_{[0,1]} |f| d\mu < \infty$, find

$$\lim_{k \rightarrow \infty} \int_{[0,1]} k \ln \left(1 + \frac{f^2}{k^2} \right) d\mu.$$

$$\lim_{k \rightarrow \infty} \int_{[0,1]} k \ln \left(1 + \frac{f^2}{k^2} \right) d\mu = \lim_{k \rightarrow 0} \int_{[0,1]} \frac{\ln(1 + f^2 k^2)}{k} d\mu$$

Using $\left| \frac{\ln(1 + f^2 k^2)}{k} \right| \leq |f|$ and DCT we can

$$\lim_{k \rightarrow 0} \int_{[0,1]} \frac{\ln(1 + f^2 k^2)}{k} d\mu = \int_{[0,1]} \lim_{k \rightarrow 0} \frac{\ln(f^2 k^2 + 1)}{k} d\mu = 0$$

10. Assume that $\{f_n\}_{n=1}^\infty, \{g_n\}_{n=1}^\infty, f, g$ are integrable and g_n are non-negative, $f_n \rightarrow f$ pointwise, $g_n \rightarrow g$ pointwise, $|f_n| \leq g_n$ a.s., and $\int g_n d\mu \rightarrow \int g d\mu$. Show that

$$\int f_n d\mu \rightarrow \int f d\mu.$$

Using Fatou lemma and the property of pointwise, we can obtain $f_n + g \geq 0, g - f_n \geq 0$,

$$\int (f + g) d\mu \leq \liminf_{n \rightarrow \infty} \int f_n d\mu + \int g d\mu \Rightarrow \int f d\mu \leq \liminf_{n \rightarrow \infty} \int f_n d\mu,$$

$$\int (g - f) d\mu = \liminf_n \int (g - f_n) d\mu \Rightarrow \int f d\mu \geq \limsup_{n \rightarrow \infty} \int f_n d\mu.$$

Then $\int f d\mu = \lim_{n \rightarrow \infty} \int f_n d\mu$.

11. Let f and $f_n, n = 1, 2, \dots$, be non-negative measurable functions on $[0, 1]$ such that $f_n \rightarrow f$ pointwise. Under each of the following additional assumptions, either prove that $\int_{[0,1]} f d\mu \rightarrow \int_{[0,1]} f_n d\mu$ or given a counter example to show that this is not generally true.

(i) $f_n \geq f$ and $\int_{[0,1]} |f_n| d\mu < \infty$ for all n .

(ii) $f_n \geq f_{n+1}$ for all n .

(iii) $f_n \leq f$ for all n .

(i) False: $f_n(x) = n1_{(0,1/n)}$ and $f = 0$.

(ii) False: $f_n(x) = \frac{1}{nx}$ and $f = 0$. (Note that $\int_0^1 f_n d\mu = \infty$ for all n .)

(iii) True. f might not be integrable, DCT is not applicable. But we can compute that

$$\int_{[0,1]} f_n d\mu \leq \int_{[0,1]} f d\mu, \quad \text{then} \quad \limsup_n \int_{[0,1]} f_n d\mu \leq \int_{[0,1]} f d\mu.$$

Moreover, by Fatou's Lemma

$$\int_{[0,1]} f d\mu = \int_{[0,1]} \liminf_n f_n d\mu \leq \liminf_n \int_{[0,1]} f_n d\mu.$$

12. Let $f : [0, 1] \rightarrow \mathbb{R}$ and $\sup \{|f(x)| : x \in [0, 1]\} \leq 1$.

(i) Show that

$$\int_{[0,1]} \sqrt{1 - f^2} d\mu \leq \sqrt{1 - \left(\int_{[0,1]} f d\mu \right)^2}$$

(ii) Describe the class of functions for which the equality takes place.

(a) By Cauchy-Schwarz inequality, we have

$$\left(\int f \right)^2 + \left(\int \sqrt{1 - f^2} \right)^2 \leq \int f^2 + \int \left(\sqrt{1 - f^2} \right)^2 = \int \left(f^2 + \left(\sqrt{1 - f^2} \right)^2 \right) = \int 1 = 1.$$

(b) Cauchy-Schwarz inequality is a special case of Hölder's inequality, hence, for strictly convex function, the equality holds if and only if,

$$cf = \sqrt{1 - f^2}, \quad f^2 = \frac{1}{c^2 + 1}, \quad c \neq 0 \text{ is a constant.}$$